Time Teaches:
The Clock as a Mathematical Object

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The hidden structure:

We are going to look at the clock.
Not at the time, but at the clock itself.

In this lecture, all positions start at 1,
not at 0.
That is a deliberate choice.
It fits how the clock itself works.
An hour hand points to 1, not to 0.
The digital root runs from 1 to 9,
not from 0 to 8.
January is month 1, not month 0.

Concretely:
the second runs from 1 to 60.
The minute from 1 to 60.
The hour from 1 to 24.
When a position is at its maximum, it stays at that maximum.
60, not 0.

That is exactly how the digital root works.
At 9 it stays 9, not 0.

The basic example

It is 10 o'clock.
You add 3 hours.
What time is it?

You can calculate it: 10 + 3 = 13.
But when you look at the clock, you do not see 13.

The hour hand points to 1.
So you say: it is 1 o'clock.

Not because 13 and 1 are the same.
But because the clock has only 12 positions.
When you go past 12, it starts again.

The position it points to, we call 1.
Yet you also know that it is 13 hours.

The clock only shows you the position.
The rest you do entirely yourself.

You write that down like this:
$13 \equiv 1 \pmod{12}$

You read this as: 13 points to the same position as 1 on a 12-hour clock.

But there is a catch.
When the clock points to position 1,

how do you know whether it is 1 o'clock at night or 1 o'clock in the afternoon?
The clock itself does not tell you.

What the clock does is make two rounds
per day.
One round during the day,
one round at night.
That is the double cycle.

Without that double cycle you would never know whether it is 01:00 or 13:00.
The double cycle is what sets the clock apart from a simple counter.
A counter is linear, not a spiral.

Position and quantity

The clock has 12 positions,
numbered from 1 to 12.
The time that has passed since the starting point is a quantity.
It can be any number:
1 hour, 13 hours, 25 hours, 37 hours.

Some quantities point to the same position.
1 hour, 13 hours, 25 hours, 37 hours
all point to position 1.
What they have in common is not the number itself, but the distance between them.
That distance is 12.

And that is why we say: the modulus is 12.

We write:
$13 \equiv 1 \pmod{12}$
$25 \equiv 1 \pmod{12}$
$37 \equiv 1 \pmod{12}$

The modulus 12 is what connects all these numbers.

How many full rounds of 12 are there in each of those numbers?
In 13 there is one.
In 25 there are two.

And two rounds of 12 hours is exactly
one full day.

The main theory

What we investigate is the structure behind the measurement of time.
That structure has a form:
$p \equiv 1 \pmod{q}$

That is the main theory.
The main theory has two forms.
One is theoretical, the other is reality.

The theoretical form is:
$q + 1 = p$.
So $p \equiv 1 \pmod{q}$, with $m = 1$.
That is one round.
You arrive at position 1.

You are one level higher.

The reality form is:
$2q + 1 = p$.
So $p \equiv 1 \pmod{q}$, with $m = 2$.
That is two rounds.
You arrive at position 1.
You are two levels higher.

The modulus stays the same.
Only the number of rounds changes.
And it is a spiral.
You keep coming back to position 1, but at a higher level.
The position is the same.
The level rises.

Two ways to read m

The letter m can be read in two ways.

Convention 1 (number of rounds):
$m = 1$: one round. $p = q + 1 = 13$. Incomplete.
$m = 2$: two rounds. $p = 2q + 1 = 25$. Complete.
Formula: $p = mq + 1$.

Convention 2 (number of cycles):
$m = 0$:
no whole cycle, but the half.
$p = q + 1 = 13$. Half.
$m = 1$:
one whole cycle.
$p = 2q + 1 = 25$. Whole.
Formula: $p = (m + 1)q + 1$.

The numbers that come out are exactly the same: 13, 25, 37, 49, ...
Only the meaning of m differs.
Convention 1 counts rounds.
Convention 2 counts cycles.

The double cycle: day and night as one unit

Because day and night together are one day, $25 \equiv 1 \pmod{12}$ does not stand for two separate rounds that you add linearly.

It stands for:
1 time day (one half)
1 time night (the other half)
Together: 1 day further.

So 25 mod 12 is one day further, not two days.

Exactly the same works for the year.
Because day-year and night-year together are one year, $25 \equiv 1 \pmod{12}$ is one year further.

And because the month has the same structure, $61 \equiv 1 \pmod{30}$ is one month further.

Just like the day (25 mod 12) and the year (25 mod 12), the month is 61 mod 30.

This is the core of the spiral: two moments (day and night) together form one unit at the next level.

The day on two moduli: 12 and 24

The day can be viewed in two ways.
Both are valid.

Mod 12 (hours per half day):
q = 12 hours.
13 ≡ 1 (mod 12): one round. Half day (12 hours).
25 ≡ 1 (mod 12): two rounds. Whole day (24 hours).
37 ≡ 1 (mod 12): three rounds. One and a half days (36 hours).

Mod 24 (hours per whole day):
q = 24 hours.
25 ≡ 1 (mod 24): one round. Whole day (24 hours).
49 ≡ 1 (mod 24): two rounds. Two days (48 hours).

You see: 25 ≡ 1 (mod 12) and 25 ≡ 1 (mod 24)
are both the whole day.
The same p-value, a different modulus.
That is the spiral.
The same position, a different level.

The process and the cycle

A process is the movement you make to go from one level to the next.
In the reality form (m = 2), a process is always two rounds.

Take the week as an example.
The process of the week: 15 ≡ 1 (mod 7).
This is one process (two rounds of 7).
And this process yields one week, but with two different moments.

If you do this process twice, you get the cycle.
The unit here is the half day (12 hours),
just like the 12-hour clock.
Two rounds are 14 half days.
That is a whole week (168 hours).

Now we come to the month.
In the calendar this is "30 days".
But mathematically, 30 (2 × 15)
is only half of the process.
To get the complete cycle of the month, we have to do the process once more: 2 × 30 = 60.

So:
15 ≡ 1 (mod 7): 1 week (1 process, 2 moments).
30: 2 weeks (2 processes, 4 moments).
This is half of the month cycle.
60: 4 weeks (4 processes, 8 moments).
This is the complete month cycle.

Note:
15, 30 and 60 are numbers of half days here.

60 half days are 30 days.
Just as on the clock we do not say that an hour has passed when only half an hour has passed.
Because 30 min is half an hour.
And 60 is 1 hour.
The whole calendar works this way.

The unit we know is twelve.
And in fact it is 12 × 2,
with two different moments.
And that double moment is m = 2.
So for the month we do not say that it is only 30.
We use 60 (4 weeks of 15) to show the completeness of the cycle.
Just as an hour has 4 × 15 minutes.

The hour that makes everything visible

It is the same pattern.
15 minutes fits 4 times into 60 minutes.
Four quarters of an hour make one hour.
4 × 14 = 56 minutes,
plus the four leftovers (4 × 1) = 60.
After that you are in the next hour,
at minute 1: 61 ≡ 1 (mod 60).

That is the formula X = 8q + 4 with q = 7.
The 8q = 56 is four times two rounds of 7.
The +4 are the four leftovers.

The month

The month operates under the parameter 61/30.

Modulus q = 30 half days.
m = 2: two rounds of 30 is 1 full month.
p = 61 ≡ 1 (mod 30).

Here 30 is the modulus (the half month).
60 is the dense cycle.
And 61 is the open cycle at the next level.

Because day-month and night-month together are one month, 61 ≡ 1 (mod 30) is one month further.

The month is 60 half days, and that is 4 × 15.

4 weeks are 4 × 14 = 56 half days = 28 days.

Four leftovers remain:
the four times that a week ends at position 1
of the next one.
That is 4 half days = 2 days.
Together: 28 + 2 = 30 days.

So those 2 days are not added separately.
They are those four leftovers together.
After day 30 you are at day 1 of
the next month.

The seasons

Four seasons of 3 months make 12 months.
Four times 90 days make 360 days.
After the fourth season you are at month 1 of the next year.

The year and the spiral of time

The year operates under the parameter 25.
Modulus q = 12 months.
m = 2: two rounds of 12.
p = 25 ≡ 1 (mod 12).

Because day-year and night-year together are one year, 25 ≡ 1 (mod 12) is one year further.

The sequence goes as follows:
25 ≡ 1 (mod 12):
1 year further.
50 ≡ 2 (mod 12):
2 years further.
75 ≡ 3 (mod 12):
3 years further.
100 ≡ 4 (mod 12):
4 years further = the leap year (366 days).

The week

A week has 7 days.
That is 14 half days (168 hours).
The 15th half day is position 1 of the next week.
That is 15 ≡ 1 (mod 7), with m = 2.

The week is a parallel cycle.

A month has 30 days.
And 30 is not a multiple of 7.
A year has 360 days.
And 360 is not a multiple of 7.
Yet the week is all around us.

The calendar counts days in months and years.
The week counts days in weeks.
They start at the same moment.
But they drift out of step.

After 7 days the week is complete.
But the calendar is still in the middle
of the first month.
After 30 days the month is complete.
But the week is in the middle of the fifth week.
After 360 days the year is complete.
But the week is again in the middle of a week.

The week and the calendar come together only when their lengths meet at the same point again.
That is after 2520 days.
And after that only after a multiple of that.

The leap year in the spiral

The year according to 360 days.
$360 \equiv 3 \pmod 7$. Remainder 3.

The leap year
is the year plus m times the remainder.
Leap year = $360 + 2 \times 3 = 366$.

Or in half days: $720 \equiv 6 \pmod 7$,
and that is the 6 extra days.
The factor 2 is m = 2:
two rounds of the remainder.

In the sequence of the year, $100 \equiv 4 \pmod{12}$
is the point where the leap year appears:
4 years further.
This is how the sequence $25 \rightarrow 50 \rightarrow 75 \rightarrow 100$
closes with the leap year.

The pattern returns at every scale:
$60 + 1 = 61$
$6 \times 61 = 366$
$366 = 360 + 6$
$6 + 1 = 7$

252 as the intersection of 60 and 12

The clock works on different scales.
Modulus 12 for the day and the year.
Modulus 60 for the minute and the month.
Now we look at the point where these two moduli come together.
That point is 252.

$4 \times 60 = 240$.
$252 - 240 = 12$.
So: $252 = 4 \times 60 + 12$.

And because $60 + 3 = 63$, 252 is also: $4 \times 63 = 252$.

Both expressions are true.
They show two sides of the same structure.

What does $4 \times 60 + 12$ mean?
60 is the modulus of the minute and the month.
$4 \times 60 = 240$ are four full rounds of 60.
The 12 is the modulus of the day and the year.
So 252 is the point where four rounds of 60 bring you exactly one modulus of 12 further.
The 12 is the bridge between the two moduli.

What does 4×63 mean?
$63 = 60 + 3$.
The 3 is the remainder of the year with respect to the week: $360 \equiv 3 \pmod 7$.
So $63 = 60 + (360 \bmod 7)$.
And so: $4 \times 63 = 4 \times (60 + 3) = 240 + 12 = 252$.

This shows that the remainder 12 can also be read as
4×3: four times the remainder of the year
with respect to the week.

The 12 that appears here is the same 12 as
the modulus of the day and the year.
The 12 is the modulus of the day and the year.
The 3 is the remainder of the year with respect to the week.

For the day: 25 ≡ 1 (mod 12),
two rounds of 12 hours = 1 day further.
For the year: 25 ≡ 1 (mod 12),
two rounds of 12 months = 1 year further.
For 252: four rounds of 60 + 12 = one modulus of 12 further.

So 252 is the point where the 60-cycle
and the 12-cycle come together.

The structure in one summary:

Expression | Meaning
4 × 60 + 12 | Four rounds of 60,
plus one modulus of 12
4 × 63 | Four times (60 + 3),
where 3 = 360 mod 7
252 | The intersection of modulus 60 and modulus 12

252 is not chosen at random.
It is the point where the 60-cycle
and the 12-cycle meet.
And where the remainder 12 is the same 12 as
the modulus of the day and the year.

The spiral in practice

We do not count the separate rounds.
We count the levels we arrive at.
Two rounds are always one whole unit.
One round is half of that.

The week:
8 ≡ 1 (mod 7):
One round.
Level 1 (half a week, 84 hours).

15 ≡ 1 (mod 7):
Two rounds (1 process).
Level 2 (1 week, 168 hours).

22 ≡ 1 (mod 7):
Three rounds.
Level 3 (a week and a half).

29 ≡ 1 (mod 7):
Four rounds (2 processes).
Level 4 (2 weeks).

The digital root:
10 ≡ 1 (mod 9):
One round.
19 ≡ 1 (mod 9):

Two rounds.
28 ≡ 1 (mod 9):
Three rounds.

The month:
31 ≡ 1 (mod 30):
One round: half a month.
61 ≡ 1 (mod 30):
Two rounds (1 process): the whole month.
91 ≡ 1 (mod 30):
Three rounds: a month and a half.

The minute:
61 ≡ 1 (mod 60):
One round: half an hour.
121 ≡ 1 (mod 60):
Two rounds: the whole hour.
181 ≡ 1 (mod 60):
Three rounds: an hour and a half.

The clock (day):
13 ≡ 1 (mod 12):
One round. Half a day.
25 ≡ 1 (mod 12):
Two rounds. A whole day.
37 ≡ 1 (mod 12):
Three rounds. A day and a half.
49 ≡ 1 (mod 12):
Four rounds. Two days.

The year:
25 ≡ 1 (mod 12):
1 year further.
50 ≡ 2 (mod 12):
2 years further.
75 ≡ 3 (mod 12):
3 years further.
100 ≡ 4 (mod 12):
4 years further = the leap year.

The position is always 1.
The level rises.
That is the spiral.

The clock as a parameter choice

The main theory is: $p \equiv 1 \pmod{q}$.
The ratio is: $m = (p - 1) / q$.

The theoretical form is $m = 1$, so $p = q + 1$.
The reality form is $m = 2$, so $p = 2q + 1$.

The clock has a specific parameter choice:
$p = 25$ and $q = 12$.
The ratio that goes with it:
$m = (25 - 1) / 12 = 2$.

The bases of the clock together form a chain: 1,000; 60; 60; 24; 7; 30; 12; 10; 10; 10.

That chain falls apart into two parts:
time and days.

The time series:
The millisecond is the smallest unit.
The second is 1,000 milliseconds.
The minute is 60 seconds.
The hour is 60 minutes.

The days series:
The day is 24 hours ($25 \equiv 1 \pmod{12}$).
The month is 30 days ($61 \equiv 1 \pmod{30}$).
The year is 12 months ($25 \equiv 1 \pmod{12}$).
The decade is 10 years ($10 \equiv 1 \pmod{9}$).
The century is 10 decades ($10 \equiv 1 \pmod{9}$).
The millennium is 10 centuries ($10 \equiv 1 \pmod{9}$).

The week is 7 days.
30 is not a multiple of 7.
That is why the week stands apart.
But the week is modularly connected with the month and the rest of the chain.

This chain determines the structure of the clock.
The cycle closes after 2,520,000 days.
That is exactly 7,000 years.

Step-by-step breakdown

Step 1. The clock
Look at a clock with a second hand.
Every tick is one second.
After 60 ticks, the minute hand moves one step.
Sixty minutes make one hour.
Every 24 hours a new day begins.
The clock divides time into fixed chunks.
Those numbers (60, 60, 24) were fixed thousands of years ago.

Step 2.
The role of the 12
The clock uses 12 in two places.
The hour hand (12 hours in a half day).
And the month (12 months in a year).
In both places the 12 is the same: a complete cycle.
The day and the year are connected by the bases 24 and 360.
The 12 is the link between the small scale (the day) and the large scale (the year).

Step 3.
Theoretical form and reality form
Theoretical form:
$m = 1$, so $q + 1 = p$. One round.
You arrive at position 1, one level higher.
Reality form:
$m = 2$, so $2q + 1 = p$. Two rounds.
You arrive at position 1,
two levels higher.
Only now is a whole unit complete.

Clock:

2(12) + 1 = 25.
Two rounds of 12 hours = 1 day.

Week:
2(7) + 1 = 15.
Two rounds of 7 = 1 week.

Digital root:
2(9) + 1 = 19.

Minute:
2(60) + 1 = 121.
Two rounds of 60 = 1 hour.

Step 4.
Every cycle has its own modulus

Clock: mod 12.
Second: mod 60.
Minute: mod 60.
Hour: mod 24.
Day of the month: mod 30.
Month of the year: mod 12.
Weekday: mod 7.
Digital root: mod 9.

The form is the same for all: $p \equiv 1 \pmod{q}$.
Theoretical: $p = q + 1$.
Reality: $p = 2q + 1$.

Step 5.
The month as 2×15
The week is $15 \equiv 1 \pmod{7}$.
The month is $30 = 2 \times 15$.
So the month is twice the p-value of
the week.
$30 \equiv 2 \pmod{7}$.
In days that is 4 weeks of 7 days
and 2 days left over.
Those 2 days are the four leftovers (4 half days).

The year: $360 = 24 \times 15$. And $360 \equiv 3 \pmod{7}$.
The decade: $3{,}600 \equiv 2 \pmod{7}$.
The century: $36{,}000 \equiv 6 \pmod{7}$.
The millennium: $360{,}000 \equiv 4 \pmod{7}$.

The whole chain is modularly connected with
the week.

Step 6.
When the clock closes
The calendar is 1,000 years of 360 days,
so 360,000 days.
The week is 7 days.
lcm(360,000; 7) = 2,520,000 days.
2,520,000 / 360 = 7,000 years.

In milliseconds: $2{,}520{,}000 \times 86{,}400{,}000 = 217{,}728{,}000{,}000{,}000$.

The full clock cycle lasts 2,520,000 days, or 7,000 years.

Step 7.
The 1,000-year cycle and the 7,000-year cycle
The 7,000-year cycle arises with a week of 7 days: lcm = 2,520,000 days = 7,000 years.
The 1,000-year cycle arises with a week of
6 days: lcm = 360,000 days = 1,000 years.
With a calendar of 360 days and a week of 7 days: lcm = 2,520 days = 7 years.
Both are valid.
It depends on which week you choose.

Step 8.
What 2,520 is
2,520 is the smallest number divisible by all numbers from 1 to 10.
It is the lcm of 1 to 10.
2520 : 7 = 360

Step 9.
What happens if you change the week
If we take 6 days: lcm(360,000; 6) = 360,000 days = 1,000 years.
The 6-day week adds nothing.
The 7-day week does.
lcm(360,000; 7) = 2,520,000 days = 7,000 years.
The week of 7 days is the only week length between 1 and 10 that does not fit into 360.
As a result the week keeps drifting out of step
with the calendar.
And that is why the clock closes only after 7 times the calendar.

Step 10.
The Gregorian cycle
The Gregorian calendar uses 365 days.
And 366 days if divisible by 4, except century years not divisible by 400.
In 400 years there are 97 leap years.
Days in 400 years: 400 × 365 + 97 = 146,097.
146,097 / 7 = 20,871.
A whole number of weeks.
The Gregorian calendar closes after 400 years.

Step 11.
Computational verification:
The mathematical structure has been tested with
a computer algorithm.

The results confirm the theory:
Zero error margin in the formulas and the palindrome structure.

Frobenius number:
with 25 and 12, all numbers above 288 can be formed.
The week (7) fits in four ways into the formula m × q + 1 = 7.
Parameter 60/61: X = 8q + 4 yields exactly 60 (at q = 7).

Cycles coincide:
the number of unique combinations of position in the millennium and weekday is exactly
2,520,000.

Spiral in practice:
adding 25 months brings you exactly to year 3, month 1 (25 ≡ 1 mod 12).

Summary

1.
The clock is a positional system with multiple bases: 1,000; 60; 60; 24; 7; 30; 12; 10; 10; 10.
Every position has its own weight.

2.
The clock has a bijection within the cycle.
There is a one-to-one mapping between N
and the clock tuple.

3.
The clock has a double cycle.
The 12-hour clock has m = 2.
The p-value of the day is 25.
Because day and night together are one day,
25 mod 12 is one day further.

4.
The clock and the calendar share the same 12. They have the same q = 12 and therefore the same p-values: 13, 25, 37, 49, ...

5.
The clock closes after 2,520,000 days, or 7,000 years.

6.
2,520 is the lcm of 1 to 10.

7.
The month is 2 × 15 = 30. And 15 ≡ 1 (mod 7). The month is modularly connected with the week.

8.
The whole chain is modularly connected with the week.

9.
The clock is a parameter choice within the general form: $p \equiv 1 \pmod{q}$ with m = 2.

10.
Two rounds are always one whole unit. One round is half of that.

11.
It is a spiral, not a sum. Every round ends at position 1 of the next level. There is no day 0, month 0 or minute 0.

12.
252 is the intersection of the 60-cycle
and the 12-cycle. 252 = 4 × 60 + 12 = 4 × 63.

13.
The day can be viewed on two moduli: mod 12 (13 = half, 25 = whole) and mod 24 (25 = whole, 49 = two days).

14.
m can be read as number of rounds or as number of cycles.
The numbers remain the same.

15.

Only 7,000 years closes,
and after that every multiple of 7,000.

16.
The Gregorian calendar closes after 400 years (146,097 days, 20,871 weeks).

17.
Computational verification confirms the theory on all fronts.

What we have done
We wrote the clock down as
a positional system
with a mixed base.
We saw how every position
has its own weight.
And how the bijection between N
and the clock tuple works.

Then we looked deeper.
The clock has 12 positions.
The day has 24 hours.
The ratio is 2.
The calendar has 12 months.
The month has 30 days.
The week has 7 days.

It is a spiral.
Every round ends at position 1
of the next level.

The full cycle lasts 2,520,000 days,
or 7,000 years.
And 2,520 is the lcm of 1 to 10.
2520:7=360

And the whole chain is modularly connected
with the week.

Only 7,000 years closes.

The clock is an instance of the main theory
$p \equiv 1 \pmod{q}$.

To the best of my knowledge, this formulation,
in which the clock is described as a parameter choice within this form, has not been written down before.
I cannot rule out that it exists elsewhere.
Anyone who knows of an earlier source is invited to share it.

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